

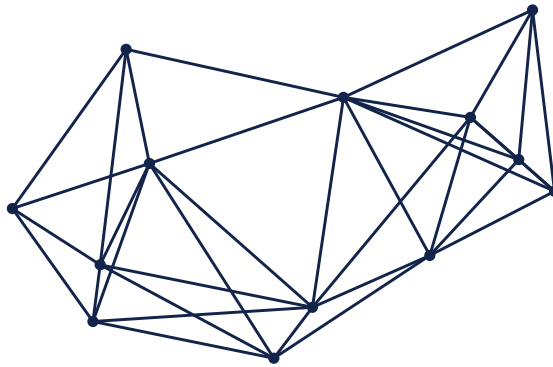
AI-POWERED CLINICAL DECISION SUPPORT • RESEARCH PROJECT BRIEF

Personalized Recommender System for Virus Research & Diagnosis Laboratory Network

Advancing Diagnostic Decision-Making through Artificial Intelligence

Amity Centre for Artificial Intelligence, Amity University, Noida, India

In partnership with ICMR – National Institute of Epidemiology, supported by the
Department of Health Research, Ministry of Health & Family Welfare, Government of India



PROJECT SUPPORTED BY THE INDIAN COUNCIL OF MEDICAL RESEARCH (ICMR)

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PROJECT OVERVIEW

Turning Clinical Symptoms into Faster, Smarter Diagnostic Decisions

India's public-health laboratories and frontline clinicians confront an unusually broad and constantly shifting spectrum of viral and infectious pathogens. When a patient presents with fever, respiratory distress, or a neurological syndrome, deciding which of dozens of possible laboratory tests to prioritise is a time-critical judgement call — one that shapes how quickly a patient is diagnosed and how efficiently a laboratory's testing capacity is used.

This project, undertaken by the Amity Centre for Artificial Intelligence in partnership with ICMR – National Institute of Epidemiology, develops an AI-based clinical decision-support system that turns a patient's demographic profile, clinical symptoms and syndromic presentation into a ranked, confidence-scored shortlist of the most probable infections — guiding the clinician toward the most relevant confirmatory laboratory test at the point of care.

DESIGNED TO SUPPORT, NOT REPLACE, CLINICAL JUDGEMENT

Every recommendation is reviewed by a treating physician and validated against confirmed laboratory results before it informs patient care.

PROJECT OBJECTIVES

**Smart Infection Triage**

Analyse patient demographics, symptoms and syndromic presentation to recommend the most probable infections and prioritise the right laboratory tests.

**Rigorous Model Optimisation**

Benchmark and refine multiple modelling approaches to maximise diagnostic accuracy and consistency with recognised clinical syndrome definitions.

**Human-in-the-Loop Validation**

Close the loop between AI recommendations, clinician review and confirmed laboratory results, so every case strengthens the system over time.

**Surveillance & Outbreak Intelligence**

Lay the groundwork to surface geographic and seasonal infection patterns across ICMR's national surveillance network, supporting early outbreak signal detection.

METHODOLOGY

From Symptom to Recommendation: How the System Works

Each case moves through a structured, six-stage pipeline that pairs rich clinical data with a purpose-built AI model — and feeds confirmed outcomes back into the system.

**1 Clinical Intake**

Patient demographics, geographic details and a structured checklist of 35 symptom indicators across neurological, respiratory, gastrointestinal, dermatological and systemic presentations are captured at the point of care.

2 Feature Engineering

Clinical, geographic (state / district) and temporal (seasonality, illness duration) signals are combined into a single structured patient profile.

3 AI Classification

A purpose-built, dual-stage deep-learning model first narrows to the most probable major infection group, then resolves closely related sub-types — each with a calibrated confidence score.

4 Syndrome Check

Recommendations are automatically cross-checked against the reported clinical syndrome, filtering out results inconsistent with the clinical picture before they reach the clinician.

5 Clinician & Lab Review

The treating physician reviews the ranked recommendations; once confirmatory laboratory results are available, they are recorded against the case.

6 Continuous Learning

Confirmed outcomes flow back into ongoing model evaluation, so the system keeps improving with every case.

TRANSPARENT BY DESIGN

Every recommendation carries a calibrated confidence score alongside its leading alternatives, so the clinician always sees the model's reasoning — never a single, unexplained answer.

SCOPE & COVERAGE

Built on One of India's Richest Clinical Surveillance Datasets

The system is trained and continuously evaluated on real-world clinical data contributed through ICMR's national surveillance network — giving it a breadth of coverage few diagnostic decision-support tools can match.

24

Major infection categories

35

Structured clinical symptom indicators

9

Recognised syndromic categories

663K+

Curated clinical cases in the training corpus

100

Pathogens in the confirmatory reference panel

35+

States & union territories represented

Recommendations span globally significant pathogens — including Dengue, Chikungunya, Japanese Encephalitis, Hepatitis A/B/C/E, Influenza A and B subtypes (H1N1, H3N2, Victoria), SARS-CoV-2, Respiratory Syncytial Virus and more — with a further refinement layer resolving additional sub-types (including Zika, West Nile virus and Kyasanur Forest Disease) within the broader 'other viral infections' group. Every recommendation is cross-referenced against ICMR's curated national reference panel of confirmatory laboratory tests, keeping the tool aligned with what laboratories can practically confirm.

CLOSING THE LOOP — FROM PREDICTION TO CONFIRMED DIAGNOSIS

Unlike a one-way prediction tool, the system tracks each case from initial AI recommendation through clinician review to confirmed laboratory outcome — building a continuously growing, ground-truthed evidence base for future refinement.

Roadmap. Extending this evidence base to surface geographic and seasonal outbreak-pattern signals across ICMR's wider national surveillance repository.

Data governance. All clinical records are de-identified and handled under access-controlled, audit-logged safeguards consistent with ICMR data-governance protocols.

INSTITUTIONAL PARTNERSHIP

A Collaboration Between AI Research and Public-Health Expertise

The project brings together specialist AI research capability with deep clinical and epidemiological expertise, under the umbrella of India's apex medical research body.



**Amity Centre for
Artificial Intelligence**

**Amity Centre for Artificial
Intelligence, Amity
University**

AI and machine-learning
research, system design and
model development.



icmr **NIE**
INDIAN COUNCIL OF
MEDICAL RESEARCH NATIONAL INSTITUTE OF
EPIDEMIOLOGY

**ICMR – National Institute
of Epidemiology**

Clinical and epidemiological
expertise, and access to India's
national viral surveillance
network.



सत्यमेव जयते
Department of
Health Research

**Department of Health
Research, Government of
India**

National public-health policy
alignment and programme
support.

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This partnership model — pairing Amity's AI research capability with ICMR's clinical and public-health authority — reflects a broader approach to translating advanced machine learning into tools that clinicians can trust and public-health laboratories can act on.

Read more about ACAI's research portfolio at amity.edu/noida/acai/research-projects.asp

OUR TEAM

The People Behind the Project

A multidisciplinary team spanning artificial-intelligence research and clinical epidemiology leads the project's day-to-day work.

PRINCIPAL INVESTIGATORS

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